SIMATS ENGINEERING

ASSESSMENT TOOL 1-Pseudocode Writing Task

COURSE NAME: R PROGRAMMING **COURSE CODE: ITA04**

COURSE OUTCOME COVERED

CO1: Apply R programming fundamentals including data structures, vectors, matrices, arrays, and class types to perform mathematical operations and manage data effectively.
(Bloom Level -BL2- BL4)

Assessment Tool Weightage: 20%

Code of Conduct:

I Pragathi KC (192321075) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Pragathi.KC.
Signature of the student:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Problem Understanding	20%	Identifies all inputs, outputs, constraints accurately	Minor missing elements	Partial understanding	Incorrect interpretation
Algorithm Design	30%	Complete, correct, and logically sound solution	Minor logical gaps	Partially correct logic	Incorrect approach
Use of Control Structures	20%	Efficient and appropriate use of loops, conditions, functions/modules	Minor inefficiencies	Limited or basic usage	Incorrect or missing constructs
Pseudocode Structure	10%	Well-structured, clear, consistent format, easy to follow	Mostly clear with minor issues	Difficult to follow in parts	Poorly structured and unclear
Completeness	20%	Handles all cases; optimized approach	Handles most cases; reasonable efficiency	Handles basic cases only	Incomplete or inefficient

96/100

PSEUDOCODE WRITING TASK

Q1. Variables and Data-types Analysis

Pseudocode

```
START
set num = 20
set name = "Deepa"
set status = TRUE

sum = num + 10
result = status and FALSE

Display num
Display name
Display status
Display sum
Display result
STOP.
```

Output

20

DEEPA

TRUE

30

FALSE

Q2. Data Structures Implementation

Pseudocode

START
Create vector = [10, 20, 30]
Create list = ["Deepi", 19 TRUE]
Create Matrix = [[1, 2][3, 4]]

Display Vector[2]
Update Vector[2] = 25
Find sum of Vector
Display Vector
Display list
Display Matrix
STOP.

Output:

Vector : 10 25 30

List
Deepi
19
TRUE

Matrix
1 2
3 4

sum = 65.

Q3. Matrix and Array Operations

PseudoCode

START

Create Matrix A

Create Matrix B

Add A and B

Multiply A and B

Create 3D array

Display Addition

Display Multiplication

Display Array

STOP.

Output:

Matrix Addition

6 8

10 12

Matrix Multiplication

19 22

43 50

3D Array Created Successfully.

Q4. Data frame Construction and Manipulation

PseudoCode

START

Create Data Frame(Name, Marks)

Add Grade Column

Delete Grade Column

Modify Marks

Display Data frame

Display Summary

STOP.

Output:

Name	Marks
Jhash	95

Q5. Functions & Logical Operation

PseudoCode

START

Create Function Square(X)

return X x X

Call Square (5)

Create Logical Vector

Perform Logical AND

Display result

STOP.

Output:

Square = 25

Logical Vector TRUE TRUE

FALSE